Multi-Object

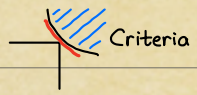
$\min f_1 \quad \min f_2$

s.t. $h = 0,\ g \le 0$

Design Space : 变量 x 的空间

Criteria Space : 函数值 f 的空间

Pareto : 左下无点



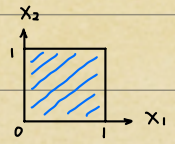
Utopia point : (x_1^*, x_2^*)

e.g. $\min f_1 = x_1 + 2x_2 \qquad \min f_2 = 3x_1 - 2x_2$

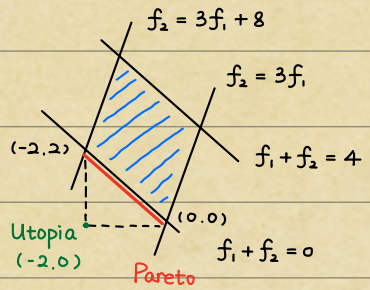
s.t. $0 \le x_1, x_2 \le 1$

$f_1 + f_2 = 4x_1 \in [0, 4]$

$3f_1 - f_2 = 8x_2 \in [0, 8]$



Design Space



Criteria Space